

YOONHAK NAM

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Profile

Quantitative researcher with a background in nuclear theory, neutron-star cooling, scientific computing, and data-driven optimization. Currently Research Lead at SPEQTRA Investment Research Pte Ltd.

Experience

SPEQTRA Investment Research Pte Ltd · Research Lead Jul. 2026 — Present

- Leading quantitative research and systematic investment analysis.

SPEQTRA Investment Research Pte Ltd · Quantitative Research Intern Jul. 2025 — Jun. 2026

- Conducted quantitative research and systematic strategy development.

Matsuo Institute, Inc. · Quantitative Research Intern Oct. 2024 — Jun. 2025

- Built Python tools for backtesting and evaluating CTA strategies.
- Implemented trading algorithms from academic and financial literature.
- Developed a research database from public data and web sources.

Publications

Y. Nam and K. Sekizawa, “Data-driven exploration of the neutron $^3\text{P}_2$ pairing gap using Cassiopeia A neutron star observational data: Direct χ^2 minimization.” *Physical Review C* 113, 045807 (2026).

Y. Nam and K. Sekizawa, “Vortex Creep Heating in Neutron Star Cooling: New Insights into Thermal Evolution of Heavy Neutron Stars.” *Proceedings of INPC2025* (2025); *arXiv:2510.24167*.

T. Lee, **Y. Nam**, and K. Sekizawa, “Impact of hyperon mixing on neutron star structure based on Skyrme-type equations of state.” *arXiv:2605.28727* (2026).

Y. Nam and K. Sekizawa, “Vortex creep heating in neutron star cooling with direct Urca processes in heavy neutron stars.” *arXiv:2511.13263* (2025).

Award

Award of Excellence · GCI 2023 Winter Feb. 2024

- Top 38 of 2,424 participants in the University of Tokyo–hosted program.
- Completed machine-learning and data-driven business projects in Python.

Education

Institute of Science Tokyo

M.S. in Physics · Nuclear Theory

GPA 4.21 / 4.50

Oct. 2024 — Jun. 2026

Neutron-star cooling, nuclear superfluidity, and data-driven optimization.

Completed M.S. 3 months ahead of schedule

Tokyo Institute of Technology

B.S. in Physics · GPA 4.17 / 4.50

Apr. 2019 — Sep. 2024

Military service: Apr. 2020 — Mar. 2022

Japan–Korea Joint Scholarship Program for Science and Engineering Students.

Completed B.S. 6 months ahead of schedule · 7 semesters

Languages

Korean · Native

Japanese · Native

JLPT N1 180/180

English · Proficient

TOEIC 960/990

Chinese · Beginner

Technical Skills

Python Fortran

Scientific Computing

Data Analysis

Interests

Quantitative Research

Market Microstructure

Physics-Inspired Modeling

High-Performance Computing